

An Automatic Rating Approach Using Machine Learning and Feature Selection for Finger Tapping in MDS-UPDRS Part III

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Abstract

We propose a machine learning-based approach with feature selection for automatic finger tapping rating in the Movement Disorder Society-Unified Parkinson's Disease Rating Scale (MDS-UPDRS) Part III assessment. A total of 160 video clips from 80 participants were used. For each video, 2D hand coordinate sequences were extracted using MediaPipe, followed by movement sequence computation, feature extraction, and selection. Clinicians provided severity scores as the gold standard. A LightGBM regressor was trained and evaluated using the resulting dataset of selected feature vectors and corresponding gold standard scores. Experimental results showed that our method achieved superior performance and demonstrated potential for reliable video-based evaluation.

Keywords—Parkinson's Disease, MDS-UPDRS, finger tapping, machine learning, feature selection, LightGBM.

1. Introduction

Parkinson's disease (PD) is a chronic neurodegenerative disorder, second only to Alzheimer's disease in prevalence [1]. It is characterized by a variety of motor

and non-motor symptoms, and its diagnosis relies on well-established clinical criteria and comprehensive neurological assessments. PD involves the progressive degeneration and loss of dopaminergic neurons in the substantia nigra of the basal ganglia, resulting in a deficiency of dopamine and other neurotransmitters in the brain[2]. The clinical manifestations of PD include hypophonia, monotonic speech, resting tremor, bradykinesia, rigidity, and postural instability, among others. These symptoms can severely impact patients' quality of life. Therefore, early screening and diagnosis, coupled with appropriate treatment under the guidance of healthcare professionals, can help slow the progression of the disease and improve patient outcomes.

The Movement Disorder Society-Unified Parkinson's Disease Rating Scale (MDS-UPDRS) is currently the most widely adopted clinical tool for assessing PD, as defined by the International Parkinson and Movement Disorder Society[3]. It is used to monitor disease progression and evaluate the severity of motor and non-motor symptoms. The MDS-UPDRS consists of four parts: Part I, II, III, and IV, each addressing different aspects of the disease. Part III is specifically designed for the clinical evaluation of motor symptoms. The clinicians assess the severity of motor impairment for each item based on the MDS-UPDRS guidelines. However, rating criteria include quantifiable and subjective com-

ponents, making it susceptible to rater variability. Islam et al. [4] indicate that even among three expert neurologists with experience in PD research, no consensus was achieved in 7.78% of the finger tapping assessment, highlighting the issue of inter-rater variability. These challenges underscore the urgent need for automated, objective rating solutions. In recent years, several approaches utilizing wearable sensors or video-based tracking have been proposed for MDS-UPDRS assessment. Wearable devices, such as inertial measurement units (IMUs), are used to detect hand movement patterns[5], while electromyography (EMG) sensors are used to measure muscular activity[6], [7]. On the other hand, video-based tracking methods employ hand pose estimation or detection frameworks, such as OpenPose[8], MediaPipe[9], and other skeletal tracking models, to extract key joint positions of the hand from 2D video recordings. Compared to wearable devices, video-based approaches offer several advantages, including contactless operation, ease of use, and the ability to conduct remote assessments without interfering with the patient's movement. Additionally, a 2024 review further highlighted the current state and future prospects of video-based hand tracking for MDS-UPDRS assessment and indicated that deep learning-based hand tracking methods can enable objective and reliable video-based evaluations[10]. Thus, this study proposes a machine learning-based approach with feature selection for the automatic rating of finger tapping in the MDS-UPDRS Part III assessment.

2. Methods

Our proposed method consists of six steps: data collection, data processing, feature extraction, feature selection, model training, and model evaluation. The details of each step are described below.

2.1. Data collection

Each participant with Parkinson's disease was instructed by clinicians to sit on a chair and perform the motor assessments from MDS-UPDRS Part III, following the official MDS-UPDRS guidelines. The entire finger tapping assessment process was recorded using a 2D video camera positioned 110 cm directly in front of the participant. Note that the finger tapping was performed starting with the right hand, followed by the left. During the assessment, clinicians evaluated the severity of each movement item based on the MDS-UPDRS guidelines and the participant's motor performance, with

severity scores ranging from 0 (normal) to 4 (severe), serving as the gold standard for subsequent analysis. Written informed consent was obtained from all participants in accordance with the protocol approved by the Institutional Review Board of Kaohsiung Medical University Hospital, Taiwan (KMUHIRB-F(I)-20210121, KMUHIRB-F(I)-20220057).

2.2. Data processing

For each finger tapping assessment video, the start and end time points of the tapping motion were manually identified and clipped through visual inspection. A hand skeleton sequence was extracted using the MediaPipe hand landmark model[11]. This process yielded the corresponding 2D coordinate sequences of 21 landmarks, denoted as $\{(x, y) | i = 0, 1, 2, \dots, 20\}$. Subsequently, a subset of these coordinates was selected for computing the physical movement sequence and performing feature extraction.

2.3. Feature extraction

For finger tapping feature extraction, we utilized the 2D coordinate sequences of hand landmarks to compute the corresponding movement sequence based on the distance between the thumb tip and the index fingertip. The movement sequence is defined as a series of values, each representing the Euclidean distance between these two landmarks in each video frame. To minimize the influence of forward or backward hand movement by the patient, the movement sequence was normalized using the Euclidean distance between the wrist and the pinky metacarpophalangeal (MCP) joint as a reference sequence. This normalization step ensured consistency in the measurement of finger tapping across frames. Subsequently, this study extracted features based on the finger tapping feature extraction method proposed by Islam et al. [4] and their publicly available source code [12]. The extracted features included motion characteristics such as amplitude, frequency, and period. These motion features were then statistically described and quantified in accordance with the MDS-UPDRS rating guidelines, to capture the trend and variability of the movement sequences. Statistical descriptors included the maximum, minimum, degree of the fitted polynomial, and among others. Finally, a total of 108 finger tapping movement features were extracted. As a result, a dataset consisting of movement feature vectors and their corresponding gold standard scores was constructed.

2.4. Feature selection

For finger tapping feature selection, we utilized ANOVA tests and Pearson's correlation coefficient (PCC) analysis to identify the most representative movement features. First, the study by Hua et al. [13] suggests that the optimal number of selected feature (K) should be proportional to the number of samples. Specifically, if the total number of participants is n , the recommended upper bound is $K \leq \sqrt{n}$. The feature selection process is shown in Figure 1. Notably, the ANOVA tests were used

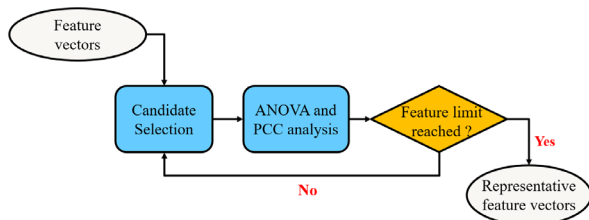


Figure 1. Flowchart of the feature selection process for movement feature vectors.

to compute the p -values of each feature across different severity levels in the dataset. Features were then ranked according to their p -values, and PCC analysis was applied to remove redundant features with high inter-feature correlation.

2.5. Model training

For finger tapping model training, we utilized stratified k -fold cross-validation to train and evaluate model performance. Subsequently, a Light Gradient Boosting Machine (LightGBM) regressor was trained and tested on each labeled dataset using the selected features. Note that the model's desired output was the severity score.

2.6. Model evaluation

For finger tapping model evaluation, we utilized three performance indices to evaluate the model's prediction outcomes. Specifically, accuracy (ACC), acceptable accuracy (AAC) [14], and the ratio of significant error (RSE) were adopted to evaluate the consistency between the model's predicted scores and the clinicians' gold standard. The three performance indices were as follows:

$$ACC = \frac{N_{|y-\hat{y}|=0}}{N} \times 100\%$$

$$AAC = \frac{N_{|y-\hat{y}|\leq 1}}{N} \times 100\%$$

$$RSE = \frac{N_{|y-\hat{y}|\geq 2}}{N} \times 100\%$$

where N represents the number of videos in the test dataset, y represents the gold standard assigned to each video, and $\hat{y}$ represents the severity score predicted by the model for each video. Note that the regression model outputs a continuous value, which is subsequently rounded to obtain the final predicted score $\hat{y}$.

3. Results

In this study, 80 participants were enrolled. Each participant performed the finger tapping task using both hands, resulting in a total of 160 finger tapping videos. The distribution of severity scores was 37.50%, 27.50%, 18.12%, 15.62%, and 1.25%, corresponding to scores of 0 to 4, respectively. Subsequently, we conducted model training and evaluation using different numbers of selected features, specifically 2, 4, 6, 8, 10, and 12 features. Experimental results demonstrated that our proposed feature selection approach achieved the best performance on the test dataset when 10 features were selected, with ACC of 41.88%, AAC of 94.38%, and RSE of 5.63%. The confusion matrix is shown in Figure 2, and the results for other feature selection combinations are listed in Table 1. In addition, to further validate the predictive performance of our proposed method, we reproduced models from previous MDS-UPDRS automatic rating studies [14], [15] and trained them on our dataset for comparison. The comparison results are listed in Table 2. Experimental results demonstrated that our model outperformed the three referenced studies across all evaluation indices.

Gold standard \ Predicted severity score	0	1	2	3	4
0	27	32	1	1	0
1	9	28	5	1	0
2	2	14	6	6	0
3	0	2	18	6	0
4	0	1	1	0	0

Figure 2. Confusion matrix of the automatic scoring model on the test set.

Table 1. Comparison of performance indices for different numbers of selected features.

Number of Feature	ACC(%)	AAC(%)	RSE
2	40.00	88.13	11.88
4	42.50	88.75	11.25
6	42.50	91.25	8.75
8	40.00	91.88	8.13
10	41.88	94.38	5.63
12	40.00	91.25	8.75

Table 2. Comparison of performance indices between our proposed method and previous studies.

Method	ACC(%)	AAC(%)	RSE
Our proposed	41.88	94.38	5.63
Yang et al. [14]	25.62	67.50	30.62
Li et al. [15]	28.75	72.50	26.87

4. Conclusion

We propose a machine learning-based approach with feature selection for the automatic rating of finger tapping in the MDS-UPDRS Part III assessment. Experimental results have demonstrated the effectiveness of our proposed method, which outperformed the three referenced studies across all evaluation indices. Therefore, it has the potential to enable objective and reliable video-based evaluations for the finger tapping task in Parkinson's MDS-UPDRS Part III assessment.

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